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XIAOHONG PENG, Guangdong Ocean University, Zhanjiang, Guangdong, China

HONGBIN JIANG, Guangdong Ocean University, Zhanjiang, Guangdong, China

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A Review of Small Object Detection Based on Deep Learning

Xiaohong Peng

College of Mathematics and Computer Science
Guangdong Ocean University
Zhanjiang, Guangdong, China
Lgdpxh@126.com

Hongbin Jiang*

College of Mathematics and Computer Science
Guangdong Ocean University
Zhanjiang, Guangdong, China
1434709585@qq.com

Abstract

In recent years, the technology of small object detection has continued to develop, resulting in numerous datasets and methods. This paper summarizes the major advancements in small object detection technology over the past decade, and analyzes 18 classic small object detection datasets. Secondly, this paper briefly describes small object detection methods, focusing on three typical methods to analyze their basic principles. Finally, this paper analyzes and forecasts the future development direction of small target detection. Future focus for small object detection should be on model architecture, attention mechanisms, multi-scale feature hierarchy, adaptive learning, and lightweight design to improve its performance and applicability.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision.

Keywords

Deep Learning, Small Object Detection, Small Object Definition

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1 INTRODUCTION

Small object detection technology has always been a key research focus in the field of computer vision. However, the development of small object detection technology has been slow due to problems such as the small pixel ratio of small objects, the difficulty in extracting features from small objects, the scarcity of relevant datasets, and the lack of standardized evaluation metrics.

With the development of deep learning technology, the field of object detection has made some progress. Deep learning models such as Faster R-CNN [1], YOLO [2] and SSD [3] have performed well in most object detection tasks. However, these techniques cannot achieve significant results in the field of small object detection. The main problem is that the pixel representation of small objects

in the image is limited, and their features are easily masked by background information, resulting in insufficient feature extraction.

To address these issues, researchers have proposed various image feature enhancement methods. For example, Feature Pyramid Network (FPN) [4] significantly improves the model's ability to detect small targets by connecting multiple scale features without increasing the computational cost of the model. DenseNet [5] proposes an aggressive feature enhancement strategy that uses dense connections to interconnect all layers to promote feature reuse, thereby significantly enhancing the feature representation of small targets.

In recent years, a variety of small object detection methods based on Transformer have emerged. DETR [6] simplifies the object detection process by removing several hand-designed components. Swin [7] solves the problem of high computational complexity in small object detection tasks by introducing a hierarchical architecture and a sliding window mechanism. DINO [8] is one of the first models to apply Transformer to solve vision problems. The authors compared the features of the original image with the features of the randomly cropped image, which significantly improved the small object detection capability. Grounded Language-Image Pre-training (GLIP) [9] combines visual and language information to further boost detection performance. RT-DETR (Real-Time Detection Transformer) [10] represents the first real-time end-to-end detection Transformer model, featuring a lightweight architecture and an efficient decoder that accelerate inference speed while preserving high accuracy, particularly for small object detection. Table 1 summarizes common small object detection methods, covering the proposers and years, core ideas, and their respective advantages and limitations.

This paper systematically summarizes the major work on small object detection. Since a clear definition of small objects is currently lacking, this paper arbitrarily defines the size range of small objects. Subsequently, representative small object detection datasets from various fields are analyzed, detailing the content and characteristics of these datasets. Based on this, the paper focuses on the challenges faced by small object detection with limited sample sizes. This paper systematically evaluates nine small object detection methods, analyzes the advantages and limitations of each method, and provides a comprehensive overview of existing solutions. Finally, this paper looks forward to the future development direction of small object detection technology from the perspectives of datasets, evaluation metrics, and the application of multimodal techniques.

* Corresponding author.



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Table 1: Summary of Common Small Object Detection Methods

Method	Proposer and Year	Core Concept	Advantages	Limitations
Faster R-CNN	Shaoqing Ren et al., 2015	Introduces Region Proposal Network (RPN) and uses ROI Pooling for fixed-size mapping.	High detection accuracy, particularly effective for medium to large-sized objects.	Performs poorly on small object detection and has slower speed.
YOLO	Joseph Redmon et al., 2016	Treats object detection as a single-stage regression problem without using region proposal methods.	Fast detection speed, suitable for real-time applications.	Small target anchor boxes are difficult to mark.
SSD	Wei Liu et al., 2016	Extracts feature maps from different layers, with each feature map responsible for detecting objects of various scales.	Primarily designed for scale-based objects, it offers high precision and strong real-time performance.	The original image was scaled up, and due to the receptive field, it became difficult to detect targets with "relatively small" size.
FPN	Tsung-Yi Lin et al., 2017	The feature pyramid is top-down, which enhances the multi-scale feature fusion of the model.	Feature pyramids can obtain sufficient image information and have a strong ability to detect small targets.	high computational complexity
FCOS	Zhi Tian et al., 2019	The anchor boxes are removed, and the boundary and category of each pixel location are predicted directly.	FPN is used for a single computation of the image to solve for bounding boxes of different sizes.	Limited ability to detect deformed objects
DETR	Carion, N et al., 2020	The Transformer architecture was used to predict object bounding boxes and labels.	The end-to-end process simplifies target detection.	The model converges slowly and has low accuracy in detecting small targets.
Swin Transformer	Liu Ze et al., 2021	It uses a layered architecture and a window self-attention mechanism to capture both local and global context.	It has high computational efficiency, strong model scalability, and achieves high accuracy in target detection tasks.	The model structure is complex, and the window size is not easy to determine.
DINO	Zhang, F et al., 2022	Based on the Transformer architecture, the query matching mechanism is optimized to enhance the target detection effect.	Even in complex scenarios, the model converges relatively quickly and maintains a relatively high level of detection accuracy.	The model requires a lot of computing resources to make predictions.
GLIP	Li, X et al., 2022	The object detection task is viewed as a context-free phrase localization task.	Utilize cross-modal pre-training to improve the understanding of images and text.	The training process is complex and requires a large dataset to achieve good results.
RT-DETR	An, Z et al., 2023	Instead of convolutional neural networks, a global self-attention mechanism is used to directly generate detection results.	Offers more efficient and faster real-time object detection performance than YOLO.	RT-DETR relies on post-processing, which increases inference latency.

2 SMALL OBJECT DATASETS AND THE DEFINITION OF SMALL OBJECTS

2.1 Small Object Datasets

The quality of datasets is crucial for research in object detection, especially in the field of small object detection. Small object detection datasets have wide applications in various aspects of human life.

In security monitoring, small object detection technology focuses on changes in video clips to achieve intelligent early warning of danger. In drone surveillance and remote sensing, detecting small objects in aerial images can infer local geological conditions and natural disasters. In medical image detection, doctors can use it to find problems in organ scans, which helps them diagnose patients. Autonomous vehicles can achieve precise obstacle avoidance in

complex traffic environments through small object recognition. In environmental protection and industrial inspection, It can even help track wildlife and prevent bad products from being sold. In sports, frame-by-frame analysis of athletes' movements and trajectories in video can guide athletes' actions and develop training programs. In astronomy, the trajectories of small celestial bodies can be tracked, thereby expanding astronomical data. Basically, spotting small stuff is a big deal in lots of areas.

2.1.1 Security Surveillance. The CUHK Avenue dataset [11] contains 16 high-resolution videos with a total of 15,328 frames. The main content is the various behaviors of students on campus. The dataset labels the location and behavior of pedestrians on campus. The ShanghaiTech dataset [12] contains as many as 330 videos with a total of 317,398 frames. It covers normal and abnormal behaviors of pedestrians in various public places on campus. The normal and abnormal behaviors are labeled accordingly, which makes it convenient for researchers to perform small target detection of abnormal behaviors on campus.

2.1.2 Drone Surveillance. The UAV123 dataset [13] has 123 videos, shot from different heights and views. It adds up to 110,000 frames. Targets such as vehicles, pedestrians, and buildings are clearly labeled in the videos. This dataset can be used for target tracking research in the field of UAV surveillance. The UAVDT dataset [14] also consists of videos taken from a high-altitude perspective, containing 82,310 video frames. Targets such as vehicles, pedestrians, and animals in complex scenes are clearly labeled, making it suitable for target detection and tracking research in UAV videos.

2.1.3 Remote Sensing. The DOTA dataset [15] has 2,806 remote sensing images. The dataset covers various ground targets such as buildings, vehicles, and ships, and provides detailed annotations for the forms required for target detection in remote sensing images. The NWPU VHR-10 dataset [16] has 800 satellite images and is also used for image analysis. It's got annotations for typical, everyday objects.

2.1.4 Medical Imaging. The LIDC-IDRI dataset [17] contains CT images of 1018 patients with pulmonary nodules. The dataset is labeled with the location of the lesions by multiple radiologists and can be used in the field of pulmonary nodule image lesion detection. The NIH Chest X-ray dataset [18] includes chest X-rays from 112,120 people who have lesions. The dataset contains 14 labels for pleural diseases. This dataset was created by the National Institutes of Health (NIH) and is suitable for target detection of lung diseases in the medical field.

2.1.5 Autonomous Driving. The KITTI dataset [19] contains 15,000 traffic scene images, of which 7,481 are used for training and 7,518 for testing. The dataset is labeled with vehicles, pedestrians and traffic signs in public transportation scenes. It can be used for object detection in autonomous driving scenarios. The Cityscapes dataset [20] contains 5,000 traffic images. The distribution of this dataset is as follows: 2,975 images are used for training, 500 for validation and 1,525 for testing. All the images are marked at the pixel level. This makes it useful for finding objects in city traffic situations.

2.1.6 Environmental Protection. The Caltech Camera Trap Dataset [21] contains 243,163 images of animal behaviors. It has detailed

notes on what all sorts of wild animals are doing, so it's great for spotting them in animal monitoring and ecological studies. The iWildCam Dataset [22] is another good one, with over 200,000 high-res pics from the field. It also has detailed notes, which help with watching animals and studying their behavior in complicated natural settings.

2.1.7 Industrial Inspection. In industrial visual inspection research, MVTec AD [23] is widely used. This dataset consists of 5,354 images of products from the production line, covering defects in 15 types of industrial products. This dataset is designed to study factory defect detection, and product defects are clearly labeled in the dataset. The DAGM dataset [24] contains 14,400 high-resolution images of industrial defects, with labels indicating the types of defects. This dataset can help control product quality and reduce the outflow rate of defective products.

2.1.8 Sports and Entertainment. The UCF101 [25] dataset contains approximately 13,320 video clips. It includes 101 motion categories. These categories cover sports activities and daily behaviors. Each motion type is clearly labeled. AVA [26] is designed for motion analysis in complex video scenes. It includes about 570,000 video clips. It is commonly used in studies on human motion detection.

2.1.9 Astronomical Observation. The SDSS dataset [27] is a dataset of over 100,000 images of celestial objects, mainly consisting of stars, galaxies, and quasars. People often use it to spot things in space. The WISE dataset [28] has millions of infrared pictures of galaxies for finding and studying infrared objects, and it has lots of notes about those objects.

This section introduces small target detection datasets for different application scenarios. Small target detection faces unique challenges in each scenario. In security monitoring, it's tough because people or cars far away can be hidden by messy backgrounds. Drone surveillance faces the problem that, against dynamic backgrounds, small targets in high-resolution images often occupy only a small number of pixels. Remote sensing images face a similar problem, which calls for more accurate target feature extraction. In medical image detection, the small details and not having much information make it tough to find tiny problems in scans. Autonomous driving requires real-time and accurate detection of fast-moving small targets in dynamic scenes. Protecting the environment by finding animals that are hiding is a hard data problem because those animals are hard to mark. Industrial inspection focuses on detecting very small defects. Astronomical observation requires identifying faint objects in complex backgrounds. Considering the characteristics of different scenarios helps small target detection perform better in real-world settings. Figure 2 shows examples of small targets in different application areas.

3 SMALL OBJECT DETECTION METHODS

3.1 Neural Network-Based Methods

At present, small object detection methods based on neural networks have gradually become a research hotspot. CNNs [29], R-CNN [30], and enhanced YOLO [31] and SSD models have been proven to have good applicability in the field of small object detection. The latest methods can further improve the performance

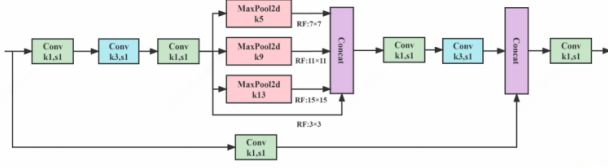


Figure 1: SPPCSPC Module Architecture Diagram

by optimizing the network architecture and training strategy of the above models. YOLOv7 [32] is a single-stage object detection model. Its speed and accuracy have been significantly improved compared with the original YOLO model, and the optimization for small object detection is particularly prominent.

3.1.1 Spatial Pyramid Pooling and Cross-Stage Partial Channels (SPPCSPC) Module. This module has two main parts: SPP and CSPNet. The SPP module [33] is used to perform pooling at multiple scales to generate feature maps of different scales. These feature maps are then concatenated to capture multi-scale information and improve detection capabilities. The CSPNet module [34] is used to split the feature maps concatenated by the SPP module. One part goes straight to the next layers, and the other part is convolved. Features from earlier layers and the convolutional layers are joined with residual connections. The channel convolution of the CSPNet module can reduce the amount of computation while enhancing feature representation.

The SPPCSPC module combines the advantages of the SPP and CSPNet modules. The SPP module enables effective detection of targets with varying sizes by introducing pooling at different spatial scales. The CSPNet module further enhances the representation of the fused features by separating and merging feature maps, making the overall design suitable for small object detection. The SPPCSPC module diagram is shown in Figure 1.

3.1.2 Coordinate Attention (CA) Module. The coordinate attention (CA) module [35] captures the link between feature map channels and orientation. It learns the dependencies between the height and width of each target map using a self-attention mechanism. Assume that, given an input feature map $X \in R^{C \times H \times W}$, it can be divided into two independent paths to process the height and width information respectively.

The module performs global average pooling along the height direction, as shown in Formula 1.

$$Zh(c, h) = \frac{1}{W} \sum_{w=1}^W X(c, h, w) \text{ for } h \in [1, H] \quad (1)$$

For the width dimension, the feature map is aggregated in the same manner. The formulation is given in Formula 2.

$$Zw(c, w) = \frac{1}{H} \sum_{h=1}^H X(c, h, w) \text{ for } w \in [1, W] \quad (2)$$

Then, 1×1 convolutions are used to restore the channel dimension. The final height A_h and width A_w attention vector is obtained through a sigmoid activation.

$$\begin{aligned} A_h &= \sigma(\text{Conv}(Zh)) \\ A_w &= \sigma(\text{Conv}(Zw)) \end{aligned} \quad (3)$$

In the end, the weights from all directions are mixed to create a CA module. This map is then put on the original feature map to change feature responses in different places. This focuses on the important details and reduces the background noise. Figure 3 shows how the CA module works.

3.1.3 Dynamic Convolution Module. The dynamic convolution module calculates different convolution kernel parameters for different input images. This effectively increases the model's capacity while keeping the number of model parameters and computational cost very small. These weights are adjusted based on the distribution of the input feature map. The dynamic convolution kernel weights are used on the feature map, which is then improved using non-linear activation and pooling layers to develop the feature representation. This module works well in areas like UAV surveillance and medical image analysis. The structure of the ODConv module is shown in Figure 4.

Table 2 (cited from K. Li et al. [36]) compares the performance of YOLOv7 with different modules, including those with the CA, SPPFCSPC, and dynamic convolution modules mentioned in the previous section. The results show that the three modules all can improve the mAP and F1 score of the results. The combination of dynamic convolution and SPPFCSPC module can achieve the best performance, with an mAP of 80.5% and an F1 score of 78.9%. Although the addition of these modules increases the model size and the number of floating-point operations, it can significantly improve the performance of small object detection.

3.2 Super-Resolution-Based Methods

Super-resolution technology is an enhancement technique that generates high-resolution (HR) images from low-resolution (LR) images, applicable to small object detection. Methods based on interpolation, edges, or deep learning can improve the detail in the original image to different extents. This technology can be applied to fields such as medical imaging, satellite remote sensing imagery, and behavioral video image analysis. Deep learning models, like SRCNN and SRGAN, can learn to turn LR images into HR images. They can generate high-quality HR images using convolutional neural networks.

However, super-resolution technology also faces many challenges in small object detection, such as motion blur and noise processing. Despite many obstacles, the continuous improvement of this method has made significant progress in the field of image processing. This paper focuses on analyzing the Super-resolution Generative Adversarial Network (SRGAN [37]). Similar to other GANs, it uses both a generator and a discriminator setup.

3.2.1 Generator Network. In the SRGAN network, the generator is an architecture consisting of B ResNet modules. In Figure 5, the k represents the kernel size, n represents the number of output channels, and s is the stride. The arrows in the figure indicate the residual structure. The label "k9n64s1" in the figure represents a 9×9 kernel with 64 output channels and a stride of 1. The function of k9n64s1 is to promote the flow of information between layers by using skip connections. It also helps prevent the gradient from disappearing in deeper networks. Structurally, SRGAN is similar to SRDenseNet. Both of them have low-level feature extraction

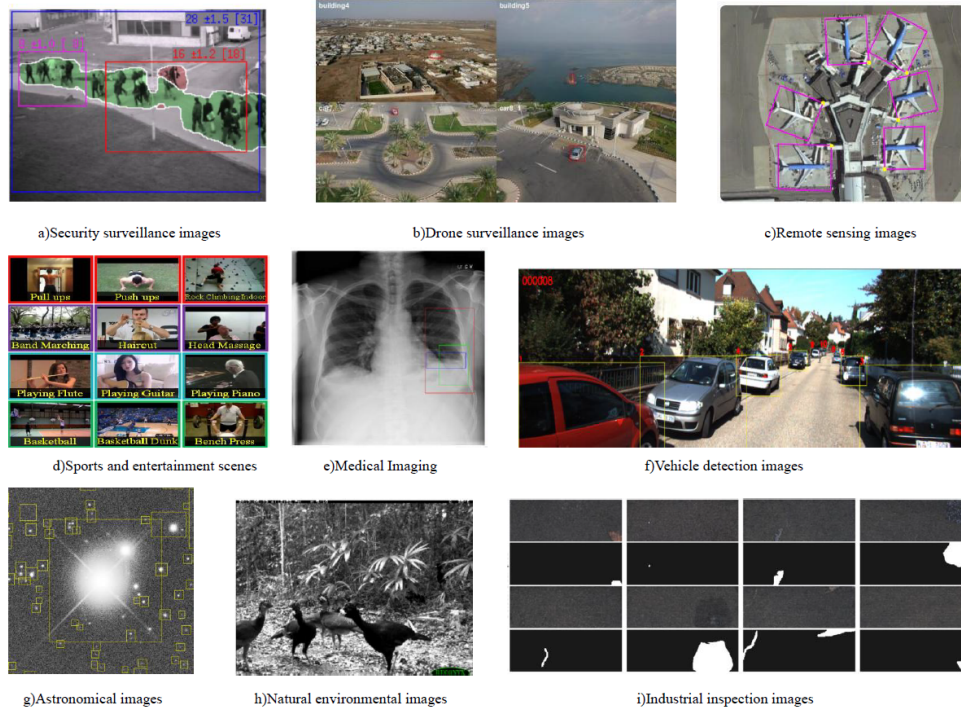


Figure 2: Examples of Small Object Detection in Different Scenarios.

Table 2: Performance Comparison of YOLOv7 with Different Modules for Small Object Detection.

Algorithm	Params(MB)	FLOPs(GB)	mAP(%)	F1-Score(%)
YOLOv7	37.6	106.5	75.9	74.2
+CA	37.7	106.7	78.8	77.3
+SPPFCSPC	37.6	106.5	78.9	77.9
+Dynamic	51.4	117.0	79.4	78.6
+CA+Dynamic	51.5	117.2	80.2	79.4
+CA+SPPFCSPC	51.5	117.2	79.5	78.3
+Dynamic+SPPFCSPC	51.5	117.2	80.5	78.9

layers, high-level feature extraction layers, deconvolution layers, and CNN reconstruction layers.

The overall flow in Figure 5 is as follows: the generator network starts from the input layer and proceeds to the initial convolutional layer for feature extraction. The network mainly uses B residual blocks with skip connections to improve feature learning. These blocks are used to enhance feature representation. The spatial dimensions of the image are recovered through upsampling layers. Finally, a convolutional layer generates the output image. The generator network can improve the quality of the generated image while preserving spatial information.

3.2.2 Discriminator Network. The SRGAN discriminator network uses convolutional and fully connected layers with nonlinear functions to tell real images from those made by the generator. It processes an input image to determine whether it is an authentic

high-resolution image or one generated by the Generator. This is accomplished through feature extraction, down-sampling, feature integration, and classification via fully connected layers.

The discriminator network in SRGAN corresponds to the maximization term in Equation 4. It uses a relatively standard CNN architecture. Leaky ReLU ($\alpha = 0.2$) [38] is used as the activation function. This helps address the issue of negative outputs. Finally, the network uses Dense blocks and sigmoid activation functions to perform binary classification, assigning scores to the ISR of the generated image and the IHR of the real image, respectively. Its basic layout mirrors the VGG [39] network. Instead, it uses stride convolution to reduce the image size and double the number of

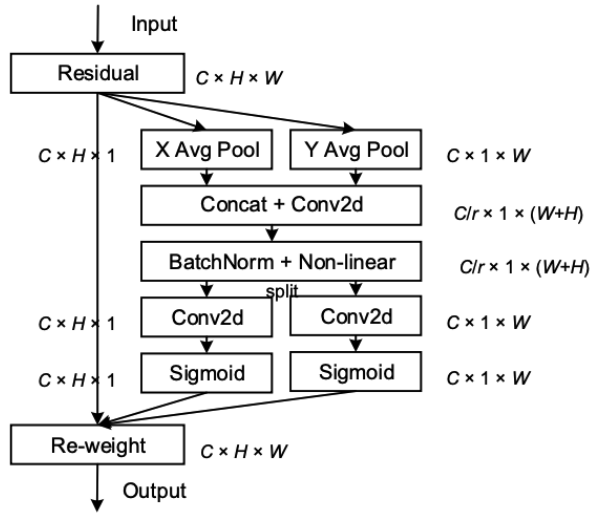


Figure 3: Coordinate Attention (CA) Module Architecture Diagram.

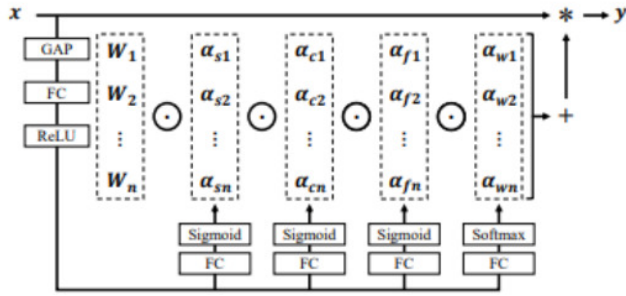


Figure 4: Dynamic Convolution Module Architecture Diagram.

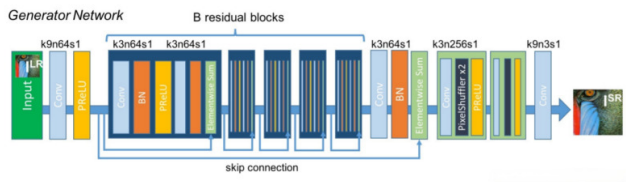


Figure 5: Generator Network Architecture Diagram.

feature maps in each layer.

$$\begin{aligned} \min_{\theta_G} \max_{\theta_D} EI^{HR} &\sim Pdata \left[\log D\theta d \left(I^{HR} \right) \right] \\ +EI^{LR} &\sim PG \left(I^{LR} \right) \left[\log(1 - D\theta d \left(G\theta g \left(I^{LR} \right) \right)) \right] \end{aligned} \quad (4)$$

The generator network is designed to transform low-resolution images into visually realistic and detailed high-resolution images. Figure 6 shows the generator and discriminator networks training against each other. In this process, the discriminator network distinguishes between generated high-resolution images and real

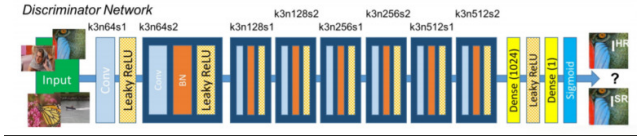


Figure 6: Discriminator Network Architecture Diagram.

high-resolution images. The generator steadily improves its performance through the repeated training process. It gradually learns to generate more realistic high-resolution images.

3.3 Methods based on Transformer

Transformer-based object detection methods also utilize their self-attention mechanism to capture global image information when inspecting small objects, thereby improving detection accuracy. The self-attention process imitates how vectors interact, getting dependencies from far away and focusing on detailed areas at image edges, which keeps small objects from being hidden by a big background. The Transformer method works with an encoder and a decoder. The encoder captures image features, and the decoder gives the detection results. The DETR framework model can be seen in Figure 7.

This diagram illustrates the fundamental structure of the DETR model. It starts with a backbone CNN that extracts a set of image features, followed by positional encoding. To incorporate positional encoding (PE) into the feature maps, sinusoidal functions are commonly used to generate the positional encodings. These encodings are added to each position in the feature map, allowing the Transformer to be aware of the position of each pixel or image patch. The formula is as follows:

$$\begin{aligned} PE(pos, 2i) &= \sin\left(\frac{pos}{10000^{2i/d}}\right) \\ PE(pos, 2i+1) &= \cos\left(\frac{pos}{10000^{2i/d}}\right) \end{aligned} \quad (5)$$

In the formula 5, pos denotes the position index, i is the feature dimension index, and d represents the dimensionality of the feature space.

Subsequently, these features are processed by a Transformer encoder, which transforms the input into an encoded image sequence. The encoder includes self-attention layers and feed-forward networks. The self-attention mechanism computes the relationships between each input element and all other elements, thereby capturing global information.

$$Attention(Q, K, V) = \text{soft max} \left(\frac{QK^T}{\sqrt{d}} \right) V \quad (6)$$

In the formula, Q , K , and V represent the query, key, and value, respectively. They are computed from the corresponding parameter matrices.

Then, the object query is processed using the Transformer decoder, which interacts with the encoded features. Finally, the prediction head uses a feedforward network (FFN) [40] to generate object category and bounding box output predictions.

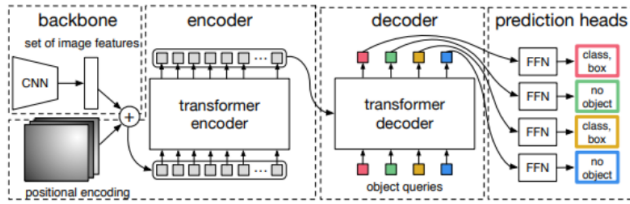


Figure 7: Basic Structure of DETR (Detection Transformer).

4 SUMMARY AND OUTLOOK

4.1 Current Challenges in Small Object Detection

Small object detection is very important in computer science and artificial intelligence, but it still has problems. To solve these problems using computers, we need better model designs and training methods. Current methods try to use better focus tools and ways to look at features at different levels to catch small details of small objects at different sizes. But, because there aren't many examples of small objects in the training data, it's hard to balance the classes, which makes the model not work as well. This shows that we need new ways to add data, re-sample data, or create fake data that is specific to the area to make the model stronger.

In the field of artificial intelligence, designing specialized algorithms for small target detection can play a crucial role in numerous areas, including autonomous vehicles, medical image analysis, and surveillance. High detection accuracy is paramount in these fields, particularly in complex and dynamic environments. Furthermore, improving the ability of AI models to detect small targets can enhance real-time applications such as robotic and unmanned aerial vehicle (UAS) systems, where the ability to detect and track small or distant targets with limited computational resources is extremely valuable.

4.2 Prospects for Small Object Detection Applications

In the field of computational science, superior model design and training methods are key to advancements in small object detection technology. Better attention mechanisms and ways to handle features at different scales should help capture small objects, no matter their size. New methods to expand the data and resampling can fix the class imbalance from limited small object training data. This can help improve model training and dependability. Using synthetic data specific to the problem can also reduce the issue of scarce small object data. It can improve how well the model works in difficult situations.

In the field of artificial intelligence applications, advancements in small object detection technology will drive development across various sectors, particularly in areas closely related to human life such as autonomous driving, medical image analysis, and surveillance systems. With the continuous development of detection algorithms, the integration of multi-layer feature fusion, intelligent data augmentation, and adaptive learning techniques will improve model accuracy and real-time performance. Furthermore, in resource-constrained environments, model compression and lightweight

design can make small object detection in applications such as unmanned aerial vehicle (UAV) systems and robotics more efficient and accurate. As these technologies mature, small object detection will provide strong support for all aspects of practical human applications, accelerating the widespread adoption of intelligent automated systems.

References

- [1] S. Ren, K. He, R. Girshick, *et al.*, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks," In *Advances in Neural Information Processing Systems (NIPS)*, 2015, pp. 1137-1149.
- [2] J. Redmon, S. Divvala, R. Girshick, *et al.*, "You Only Look Once: Unified, Real-Time Object Detection," In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Las Vegas, NV, USA, 2016.
- [3] W. Liu, D. Anguelov, D. Erhan, *et al.*, "SSD: Single Shot MultiBox Detector," In *European Conference on Computer Vision (ECCV)*, 2016, pp. 21-37.
- [4] T. Y. Lin, P. Dollar, R. Girshick, *et al.*, "Feature Pyramid Networks for Object Detection," In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, Honolulu, HI, 2017.
- [5] Z. Tian, C. Shen, H. Chen, *et al.*, "FCOS: Fully Convolutional One-Stage Object Detection," In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, Seoul, Korea (South), 2019.
- [6] N. Carion, F. Massa, G. Synnaeve, *et al.*, "End-to-End Object Detection with Transformers," In *Computer Vision – ECCV 2020, Lecture Notes in Computer Science*, 2020, pp. 213-229.
- [7] Z. Liu, Y. Lin, Y. Cao, *et al.*, "Swin Transformer: Hierarchical Vision Transformer using Shifted Windows," In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, Montreal, QC, Canada, 2021.
- [8] F. Li, *et al.*, "Mask DINO: Towards A Unified Transformer-based Framework for Object Detection and Segmentation," In *2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2022*, pp. 3041-3050.
- [9] L. H. Li, P. Zhang, H. Zhang, *et al.*, "Grounded Language-Image Pre-training," In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, New Orleans, LA, USA, 2022.
- [10] Y. Zhao, W. Lv, S. Xu, *et al.*, "RT-DETR: Real-Time Detection Transformer" [J]. *arXiv preprint arXiv:2304.08069*, 2023.
- [11] C. Lu, J. Shi, and J. Jia, "Abnormal Event Detection at 150 FPS in MATLAB," In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, 2013, pp. 2720-2727.
- [12] W. Liu, W. Luo, D. Lian, and S. Gao, "Future Frame Prediction for Anomaly Detection – A New Baseline," In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 6536-6545.
- [13] M. Mueller, N. Smith, and B. Ghanem, "A Benchmark and Simulator for UAV Tracking," In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2016, pp. 445-461.
- [14] J. Du, L. Zhao, Y. Wen, Y. Bian, G. Wang, and Z. Hu, "The Unmanned Aerial Vehicle Benchmark: Object Detection and Tracking," In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 375-391.
- [15] X. Xia, Z. Bai, G.-S. Xia, *et al.*, "DOTA: A large-scale dataset for object detection in aerial images," In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 3974-3983.
- [16] G. Cheng, J. Han, X. Lu, "Remote Sensing Image Scene Classification: Benchmark and State of the Art," In *Proceedings of the IEEE*, 2017, pp. 1865-1883.
- [17] S. Armato III, G. McLennan, L. Bidaut, M. McNitt-Gray, C. Meyer, *et al.*, "The Lung Image Database Consortium (LIDC) and Image Database Resource Initiative (IDRI): A completed reference database of lung nodules on CT scans," *Medical Physics*, vol. 38, no. 2, 2011, pp. 915-931.
- [18] X. Wang, Y. Peng, L. Lu, Z. Lu, M. Bagheri, and R. M. Summers, "ChestX-ray8: Hospital-scale chest X-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases," In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 2097-2106.
- [19] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? The KITTI vision benchmark suite," In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2012, pp. 3354-3361.
- [20] M. Cordts, M. Omran, S. Ramos, *et al.*, "The Cityscapes Dataset for Semantic Urban Scene Understanding," In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 3213-3223.
- [21] S. Beery, G. Van Horn, and P. Perona, "Recognition in Terra Incognita," In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2018, pp. 472-489.
- [22] M. McCoy, S. Beery, S. W. Cole, and R. S. Nelson, "iWildCam 2020 Competition Dataset," *arXiv preprint arXiv:2008.07273*, 2020.
- [23] P. Bergmann, M. Fauser, D. Sattlegger, and C. Steger, "MVTec AD - A Comprehensive Real-World Dataset for Unsupervised Anomaly Detection," In *Proceedings*

- of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2019, pp. 9592-9600.
- [24] W. Fauser, D. Westermann, and C. Kondermann, "A Benchmark for Industrial Optical Inspection," in Proceedings of the DAGM Symposium, 2011, pp. 262-271.
 - [25] K. Soomro, A. R. Zamir, and M. Shah, "UCF101: A dataset of 101 human actions classes from videos in the wild," arXiv preprint arXiv:1212.0402, 2012.
 - [26] C. Gu, C. Sun, D. A. Ross, *et al.*, "AVA: A video dataset of spatio-temporally localized atomic visual actions," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2018, pp. 6047-6056.
 - [27] K. N. Abazajian *et al.*, "The Seventh Data Release of the Sloan Digital Sky Survey," The Astrophysical Journal Supplement Series, vol. 182, no. 2, 2009, pp. 543-558.
 - [28] E. L. Wright *et al.*, "The Wide-field Infrared Survey Explorer (WISE): Mission Description and Initial On-orbit Performance," The Astronomical Journal, vol. 140, no. 6, 2010, pp. 1868-1881.
 - [29] A. Krizhevsky, I. Sutskever, G. E. Hinton, "ImageNet Classification with Deep Convolutional Neural Networks," Communications of the ACM, 2017, pp. 84-90.
 - [30] R. Girshick, J. Donahue, T. Darrell, and J. Malik, "Rich Feature Hierarchies for Accurate Object Detection and Semantic Segmentation," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition, 2014, pp. 580-587.
 - [31] C. Y. Wang, A. Bochkovskiy, and H. Y. M. Liao, "YOLOv7: Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors," arXiv preprint arXiv:2107.08430, 2021.
 - [32] C. Y. Wang, A. Bochkovskiy, and H. Y. M. Liao, "YOLOv7: Trainable Bag-of-Freebies Sets New State-of-the-Art for Real-Time Object Detectors," arXiv preprint arXiv:2207.02696, 2022.
 - [33] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial Pyramid Pooling in Deep Convolutional Networks for Visual Recognition," in European Conference on Computer Vision (ECCV), 2014.
 - [34] C. Y. Wang, H. Y. Mark Liao, Y. H. Wu, *et al.*, "CSPNet: A New Backbone that can Enhance Learning Capability of CNN," in 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), Seattle, WA, USA, 2020.
 - [35] Q. Hou, D. Zhou, and J. Feng, "Coordinate Attention for Efficient Mobile Network Design," in 2021 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), Nashville, TN, USA, 2021.
 - [36] K. Li, Y. Wang, and Z. Hu, "Improved YOLOv7 for Small Object Detection Algorithm Based on Attention and Dynamic Convolution," Applied Sciences, vol. 13, no. 16, 2023.
 - [37] C. Ledig, L. Theis, F. Huszar, *et al.*, "Photo-Realistic Single Image Super-Resolution Using a Generative Adversarial Network," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017.
 - [38] A. Radford, L. Metz, and S. Chintala, "Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks," Cornell University - arXiv, 2015.
 - [39] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," in International Conference on Learning Representations (ICLR), 2015.
 - [40] A. Vaswani, N. Shazeer, N. Parmar, *et al.*, "Attention is All You Need," in Advances in Neural Information Processing Systems (NeurIPS), 2017.